
Analog Electronics Lab

Laboratory Manual

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Preface

This manual has been prepared to accompany the laboratory course **Analog Electronics Lab (PCEEL308)(2024 scheme)**, offered in Semester 3 of the B.Tech. programme in Electrical and Electronics Engineering. The course is designed to provide hands-on exposure to the design, simulation, and hardware realisation of fundamental analog electronic circuits employing diodes, BJTs, JFETs, MOSFETs, operational amplifiers, and the 555 timer IC.

Each experiment in this manual follows a uniform structure consisting of the aim, theoretical background, pre-lab assignment, components and equipment required, circuit diagram, design procedure, experimental procedure, expected/model graphs, observation tables, results, and viva-voce questions. Students are expected to complete the pre-lab assignment prior to each laboratory session and to record all readings directly in the laboratory record during the conduct of the experiment.

Course Information

Course Objectives

1. To design transistor and operational-amplifier based electronic circuits.
2. To simulate and implement the designed circuits in hardware and validate experimental results against theoretical/simulated predictions.

Course Outcomes

On successful completion of this course, the student will be able to:

CO	Course Outcome	Bloom's Level
CO1	Use the various electronic instruments for conducting experiments.	K3
CO2	Design and develop various electronic circuits using diodes and Zener diodes.	K3
CO3	Design and implement amplifier and oscillator circuits using BJT and JFET.	K3
CO4	Design and implement basic circuits using IC (OPAMP and 555 timer).	K3
CO5	Simulate electronic circuits using circuit simulation software.	K3
CO6	Use PCB layout software for circuit design.	K3

Table 1: Course Outcomes and Bloom's Knowledge Level

K1: Remember, K2: Understand, K3: Apply, K4: Analyse, K5: Evaluate, K6: Create

Scheme of Evaluation

Continuous Internal Evaluation (CIE – 50 Marks)

Component	Marks
Attendance	5
Preparation / Pre-Lab Work, Viva, and Timely Completion of Lab Reports / Record (Continuous Assessment)	25
Internal Examination	20
Total	50

Table 2: CIE Marks Distribution

The continuous assessment component (25 marks) is further broken down as:

- **Preparation and Pre-Lab Work (7 Marks):** Pre-lab assignments/quizzes and understanding of theoretical background.
- **Conduct of Experiments (7 Marks):** Adherence to procedure, accurate execution, safety, skill proficiency, and teamwork.
- **Lab Reports and Record Keeping (6 Marks):** Clarity, completeness, accuracy, and timely submission.
- **Viva Voce (5 Marks):** Ability to explain the experiment, results, and underlying principles.

The final marks for preparation, conduct, viva, and record are computed as the average across all experiments listed in the syllabus.

End Semester Examination (ESE – 50 Marks)

Component	Marks
Procedure / Preliminary Work / Design / Algorithm	10
Conduct of Experiment / Execution of Work / Troubleshooting / Programming	15
Result with Valid Inference / Quality of Output	10
Viva Voce	10
Record	5
Total	50

Table 3: ESE Marks Distribution

- Students shall be permitted to appear for the end-semester examination only upon submission of a duly certified record.
- The external examiner shall endorse the record at the time of the examination.

CO–PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	2	3	3	3	3				3	3		
CO3	2	3	3	3	3				3	3		
CO4	2	3	3	3	3				3	3		
CO5	2	3	3	3	3				3	3		
CO6	3								3	3		

Table 4: CO–PO Mapping (1: Slight, 2: Moderate, 3: Substantial, blank: No Correlation)

Text Books and References

1. Robert T. Paynter, *Introductory Electronic Devices and Circuits*, Pearson Education.
2. Boylestad R. L. and Nashelsky L., *Electronic Devices and Circuit Theory*, Pearson Education.
3. Donald A. Neamen, *Electronic Circuits: Analysis and Design*, McGraw Hill.

General Instructions to Students

Laboratory Conduct

1. Students must bring the laboratory record, observation notebook, graph sheets, and a scientific calculator to every session.
2. The pre-lab assignment for the experiment scheduled for the day must be completed and ready for evaluation before entering the laboratory.
3. Students shall enter the laboratory only with permission and shall occupy the workbench allotted to them.
4. Circuit connections must be checked and approved by the faculty-in-charge or laboratory instructor before the power supply is switched on.
5. Readings must be entered directly into the observation notebook at the time of the experiment; fabricated or estimated readings will not be accepted.
6. Calculations, graphs, and inferences must be completed and the experiment got corrected on the same day, wherever possible, or within the stipulated time announced by the instructor.
7. Students must simulate the circuit using the prescribed circuit simulation software prior to hardware implementation, wherever applicable.
8. Mobile phones shall be kept in silent mode and shall not be used during the conduct of the experiment, except as directed for simulation work.
9. Any damage to equipment or components due to negligence shall be reported immediately and may be subject to the institution's equipment-damage policy.
10. Students are responsible for returning all components, probes, and patch cords to their designated locations and for switching off all instruments and the workbench power supply before leaving the laboratory.

Lab Record Submission

1. The fair record must contain, for each experiment: aim, components/equipment list, circuit diagram, design procedure, experimental procedure, observation table, model graph, calculations, result, and inferences.
2. Records must be certified by the faculty-in-charge within one week of completion of the experiment.
3. Submission of a duly certified record is mandatory for eligibility to appear for the end-semester examination.

Evaluation of Pre-Lab Assignments

Two pre-lab assignments are prescribed by the syllabus prior to the regular experiments:

1. Measurement of current, voltage, frequency, and phase shift of a signal in an RC network using a cathode-ray oscilloscope (CRO) / digital storage oscilloscope (DSO).
2. Introduction to circuit simulation using a circuit simulation software package (e.g., LTspice, PSpice, Multisim, or NgSpice/QUCS).

Students are expected to complete these foundational exercises before attempting the numbered experiments listed in [Table 6](#).

Safety Guidelines

Safety Note

1. Always verify the power supply voltage and polarity before connecting it to the circuit. Reversed polarity can permanently damage diodes, transistors, op-amps, and ICs.
2. Switch off the power supply before making, modifying, or removing any circuit connections.
3. Do not exceed the maximum voltage, current, or power ratings of any active or passive component. Always compute and verify current-limiting resistor values before energising a circuit.
4. Use current-limiting resistors in series with LEDs, diodes, and Zener diodes during testing to prevent excessive current flow.
5. Avoid short-circuiting the output terminals of a regulated DC power supply or the output of an operational amplifier.
6. Handle CRO/DSO probes with care; ensure the ground (common) lead of the probe is connected to the circuit ground/common point to avoid ground loops and inaccurate measurements.
7. Be aware of the heat-sink and case temperature of power transistors and voltage regulators during prolonged operation; avoid direct contact.
8. Electrolytic capacitors are polarised. Verify polarity before connecting; reversed polarity may cause the capacitor to overheat, vent, or rupture.
9. Do not work on live circuits with wet hands or in a wet environment.
10. In the event of a component overheating, emitting smoke, or unusual odour, immediately switch off the power supply and inform the laboratory instructor before attempting further investigation.
11. Familiarise yourself with the location of the mains isolation switch and the fire extinguisher in the laboratory.

Precautions

1. Identify the pin configuration (pinout) of every IC and transistor from its datasheet before inserting it into the breadboard.
2. Avoid loose or floating connections, particularly at the input terminals of operational

amplifiers, since floating CMOS/op-amp inputs may result in unpredictable circuit behaviour.

3. Keep signal leads and connecting wires as short as practicable to minimise stray capacitance and inductance, particularly in high-frequency oscillator circuits.
4. Re-verify all connections against the circuit diagram before switching on the power supply, and recheck after every modification.

List of Equipment and Software

Equipment / Software / Components	Typical Use in this Lab
Regulated DC Power Supply (dual/triple output)	Biasing of transistor, op-amp, and 555 timer circuits
Function / Signal Generator	Input excitation for amplifiers, filters, and frequency-response studies
Digital Storage Oscilloscope (DSO)	Measurement of waveforms, amplitude, frequency, and phase shift
Digital Multimeter (DMM)	DC bias point and resistance/voltage/current measurements
Breadboard and connecting wires	Circuit prototyping
Active and Passive Components (diodes, Zener diodes, BJTs, JFETs, MOSFETs, resistors, capacitors, inductors)	Circuit construction across experiments
Operational Amplifier ICs (e.g., μ A741, LM358)	Op-amp based experiments
Comparator IC (e.g., LM311)	Comparator and Schmitt trigger experiments
555 Timer IC	Astable and monostable multivibrator experiments
Circuit Simulation Software (e.g., LTspice / Qspice / NgSpice-QUCS)	Pre-implementation simulation of all designed circuits
PCB Layout Software (e.g., KiCad)	Introduction to PCB design

Table 5: Indicative list of equipment and software used in the course

List of Experiments

Table 6: List of Experiments – Analog Electronics Lab (PCEEL308)

Expt. No.	Title	CO
PL-1	Measurement of current, voltage, frequency, and phase shift of a signal in an RC network using a CRO/DSO (Pre-Lab)	CO1
PL-2	Introduction to circuit simulation using a circuit simulation software (Pre-Lab)	CO5
1	Clipping and clamping circuits using diodes	CO2
2	Basic RC circuits – High-pass and Low-pass filters	CO1
3	RC-coupled amplifier using BJT in CE configuration – Measurement of gain, bandwidth, and plotting of frequency response	CO3
4	Emitter follower amplifier	CO3
5	JFET amplifier – Measurement of gain, bandwidth, and plotting of frequency response	CO3
6	MOSFET amplifier	CO3
7	Design and testing of voltage regulators – Zener and series	CO2
8	Design and setup of inverting and non-inverting amplifiers	CO4
9	Op-amp circuits – Scale changer, adder, integrator, and differentiator	CO4
10	Precision rectifier using op-amp	CO4
11	Op-amp oscillators – RC phase-shift and Wien bridge oscillator	CO4
12	Op-amp oscillator – LC oscillator (Colpitts or Hartley)	CO4
13	Waveform generation – Square, triangular, and sawtooth waveform generation using op-amps	CO4
14	Basic comparator and Schmitt trigger circuits using op-amp (using comparator ICs such as LM311)	CO4
15	Active filters (one each of high-pass and low-pass)	CO4
16	Instrumentation amplifier	CO4
17	Astable and monostable multivibrator circuits using 555 timer IC	CO4
18	Introduction to PCB layout software	CO6

Format Followed for Each Experiment

Each experiment in the chapters that follow is presented under the following uniform headings:

1. **Aim**
2. **Pre-Lab Assignment**

3. **Components and Equipment Required**
4. **Theory**
5. **Circuit Diagram**
6. **Design**
7. **Procedure**
8. **Precautions**
9. **Model Graph / Expected Waveform**
10. **Observation Table**
11. **Result**
12. **Questions**

Pre-Lab 1

Measurement of Voltage, Frequency and Phase Shift in an RC Network using a Digital Storage Oscilloscope

Objectives

- (i) To become familiar with the front-panel controls of a basic (20–50 MHz) Digital Storage Oscilloscope (DSO) and a function generator.
- (ii) To measure the amplitude (peak, peak-to-peak, and RMS voltage) and frequency of a sinusoidal signal using the DSO.
- (iii) To study the response of a simple series RC network to a sinusoidal input.
- (iv) To measure the phase shift introduced by the RC network between the input and output voltages, using both the **time-difference method** and the **Lissajous (X–Y) method**.
- (v) To compare the experimentally measured phase shift and gain with the theoretically predicted values.

Pre-Lab Theory

The Series RC Network

Consider the series RC circuit shown in Figure 1, driven by a sinusoidal source $v_{in}(t) = V_m \sin(\omega t)$, where the output is taken across the capacitor. This simple linear, passive, circuit has several practical applications. This circuit is often called an *integrator* or a *low-pass filter*. The circuit when given a sinusoidal signal input, outputs a sinusoidal signal, but with different amplitude and phase, depending on the values of R and C .

The transfer function¹ of this network is

$$H(j\omega) = \frac{V_{out}}{V_{in}} = \frac{1/j\omega C}{R + 1/j\omega C} = \frac{1}{1 + j\omega RC}. \quad (1)$$

It may be noted that this can be obtained by sinusoidal analysis of the network considering R and C as impedances, and using voltage-divider formula.² Defining the *cutoff (corner) frequency*

$$f_c = \frac{1}{2\pi RC}, \quad (2)$$

¹The idea of transfer function is introduced in the course Circuits and Networks. In simple terms, it is the ratio of output to input

² $V_{Z_2} = V_{in} \frac{Z_2}{Z_1 + Z_2}$

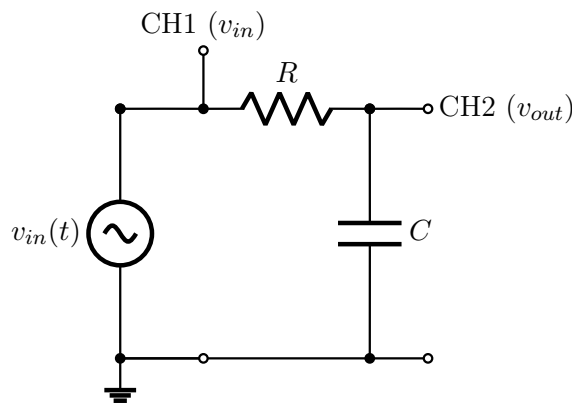


Figure 1: Series RC low-pass network. CH1 of the DSO monitors the source voltage $v_{in}(t)$ and CH2 monitors the voltage across the capacitor $v_{out}(t)$. Both channels share a common ground, which is same as the reference ground of the source.

the magnitude and phase of $H(j\omega)$ are

$$|H(j\omega)| = \frac{1}{\sqrt{1 + (f/f_c)^2}}, \quad (3)$$

$$\phi(f) = -\tan^{-1}\left(\frac{f}{f_c}\right). \quad (4)$$

The cut-off frequency f_c is an important parameter that decides the operating characteristics of this circuit. The circuit behaves differently to signals with frequencies above and below f_c .

Equation (3) shows that the network attenuates the input as frequency increases (a low-pass filter), while Equation (4) shows that $v_{out}(t)$ lags $v_{in}(t)$ by an angle ϕ that increases from 0° (at $f \ll f_c$) to 90° (at $f \gg f_c$), with $\phi = -45^\circ$ exactly at $f = f_c$.

Measurement Procedure

The circuit is set-up on a bread-board. The input voltage $v_{in}(t)$ is given from a **Function Generator**. The control knobs in the front-panel of the function generator can set the amplitude, function (sine, triangle, rectangular, square waveforms), frequency, and DC offset. Before applying the input to the circuit, observe the waveform on a Digital Storage Oscilloscope (DSO) to ensure that the input signal is as per the requirements (amplitude, form, frequency and offset).

Relating Time Delay to Phase Shift

On a DSO, phase shift is most directly obtained from the time delay Δt between corresponding zero crossings of $v_{in}(t)$ and $v_{out}(t)$, together with the period T of the waveform:

$$\phi(\text{in degrees}) = \frac{\Delta t}{T} \times 360^\circ = \Delta t f \times 360^\circ. \quad (5)$$

See the figure:

Care must be taken to note *which* channel leads and which lags, since Equation (5) gives only the magnitude of ϕ unless signs are tracked from the oscilloscope trace.

The Lissajous Method

If the DSO is operated in X-Y mode (CH1 on the horizontal axis, CH2 on the vertical axis), two sinusoids of the same frequency but different phase produce an elliptical trace (a *Lissajous*

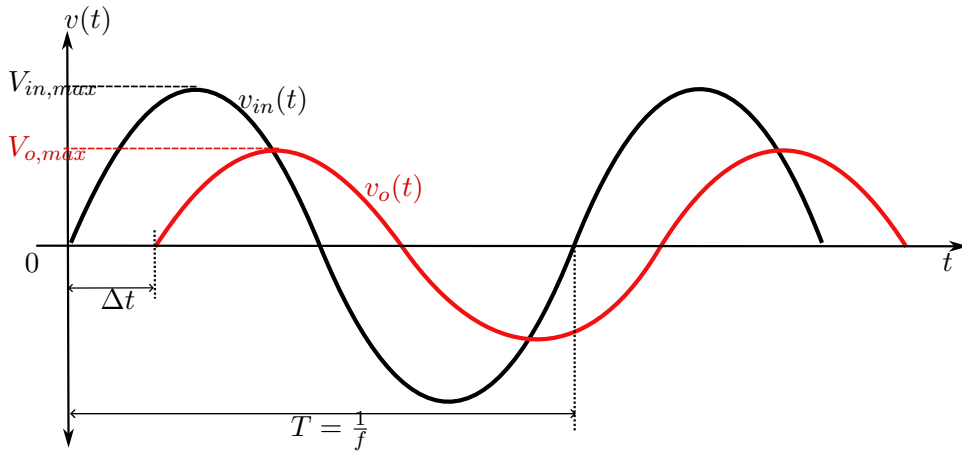


Figure 2: Input and output waveforms and salient parameters

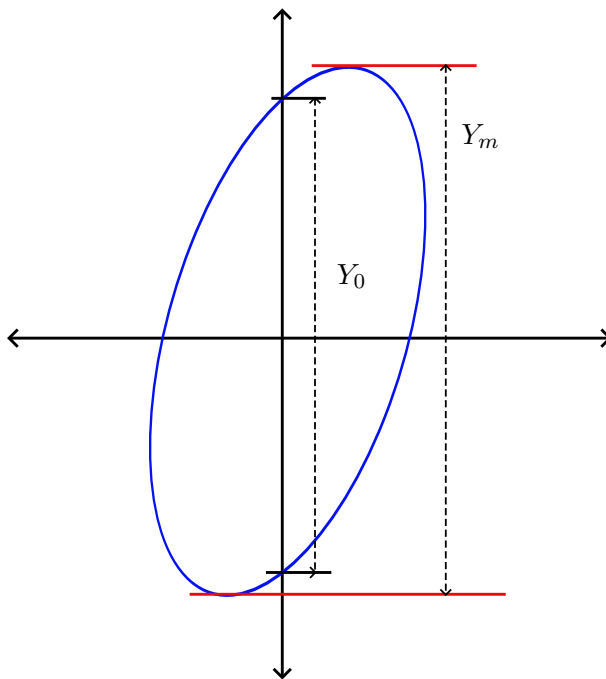


Figure 3: Lissajous pattern and phase-shift calculation

figure)³. See figure 3. Y_0 is the distance between the y-intercepts and Y_m is the distance between the extreme vertical points. The phase shift is then given by:

$$\phi = \sin^{-1} \left(\frac{Y_0}{Y_m} \right). \quad (6)$$

This method is a useful independent cross-check of the time-difference method and is more accurate near $\phi = 90^\circ$, where Δt becomes difficult to resolve on the time-domain trace.

Note

Modern DSOs have advanced measurement features that enable direct reading of the phase difference between two waveforms on screen, in addition to cursor measurements. For more details, consult the operating manual of the DSO model.

³You may explore [this online](#) simulation to learn more about Lissajous patterns.

Apparatus Required

S.No.	Equipment	Specification
1	Digital Storage Oscilloscope	2-channel, 20–50 MHz bandwidth
2	Function Generator	Sine wave, 1 Hz – 1 MHz
3	Resistor	1 (value assigned in lab, e.g. 1 k Ω)
4	Capacitor	1 (value assigned in lab, e.g. 0.1 μ F)
5	Breadboard and connecting wires	–
6	BNC–BNC and BNC–crocodile probes	As required

Table 7: Equipment list.

Pre-Lab Preparation (to be completed *before* coming to the laboratory)

Given the values of R and C assigned to your batch:

- Calculate the cutoff frequency f_c using Equation (2).
- Using Equations (3) and (4), complete the theoretical prediction table below for input amplitude $V_m = 1$ V (peak), at frequencies referenced to f_c .
- Sketch, on graph paper, the expected shape of the Lissajous figure at $f = 0.1f_c$, $f = f_c$, and $f = 10f_c$.

f	f/f_c	$ H(j\omega) $	$V_{out,pp}$ (theory)	ϕ (theory, deg)	Δt (theory)
$0.1f_c$	0.1				
$0.5f_c$	0.5				
f_c	1				
$2f_c$	2				
$10f_c$	10				

Table 8: Theoretical prediction table (to be filled in during pre-lab preparation).

Experimental Procedure (Overview)

- Connect the series RC network as shown in Figure 1. Set the function generator to produce a sine wave of the specified amplitude and set its frequency to $0.1f_c$. Make sure that the output has no DC offset, i.e., the output is a pure ac sine wave.
- Connect DSO CH1 across the function generator output (v_{in}) and CH2 across the capacitor (v_{out}), ensuring a *common ground* between the generator, the circuit, and both probes.
- Use AUTOSet (or manual Volts/Div and Time/Div adjustment) to display 2–3 complete cycles of both waveforms on screen.
- Using the DSO's horizontal cursors, measure the peak-to-peak voltage $V_{in,pp}$ and $V_{out,pp}$ for each channel.
- Using the vertical (time) cursors, measure the period T of the waveform and hence the frequency $f = 1/T$; cross-check against the function generator's dial/display setting.

6. Using the time cursors, measure the time delay Δt between corresponding zero crossings of v_{in} and v_{out} and calculate ϕ from Equation (5).
7. Switch the DSO to **X–Y mode** and record the resulting Lissajous figure. Measure Y_0 and Y_m and calculate ϕ from Equation (6).
8. Repeat Steps 4–7 for each frequency listed in Table 8 (and any additional frequencies specified by the instructor).
9. Tabulate all measured quantities alongside the theoretical predictions and calculate the percentage deviation.

Precautions

- Ensure the ground leads of the function generator and both oscilloscope probes are connected to a common reference node to avoid ground-loop errors.
- Compensate the oscilloscope probes (using the built-in calibration signal) before taking measurements. You may use the probe calibration facility of the DSO.
- Keep the function generator output amplitude constant while varying frequency; check it does not change due to output loading.⁴
- Use the $\times 1$ or $\times 10$ probe setting consistently and account for the attenuation factor in the DSO's channel settings.

Practical Observations

Append the table 8 with additional columns of practical observations. Provide your inferences and conclusions/results.

Pre-Lab Questions

- Q1. Derive Equation (1) from first principles using the voltage divider rule in the phasor domain.
- Q2. Show that at $f = f_c$, the output amplitude is $1/\sqrt{2}$ times the input amplitude and the phase shift is exactly -45° .
- Q3. If the output were taken across R instead of C , what type of filter results? Derive the new expressions for $|H(j\omega)|$ and $\phi(f)$.
- Q4. Why is the Lissajous method more reliable than the time-delay method for measuring phase shifts close to 90° ?
- Q5. List two sources of error in measuring Δt directly from the DSO screen and suggest how each may be minimized.

⁴Loading happens when the Circuit Under Test (CUT) has low input impedance compared to the function generator's output impedance.

Pre-Lab 2: Introduction to Circuit Simulation using LTspice

Objectives

- (i) To install and set up LTspice, a free SPICE-based analogue circuit simulator.
- (ii) To become familiar with the basic workflow of schematic capture, simulation setup, and waveform viewing in LTspice.
- (iii) To simulate a simple RC network and verify the results against hand-calculated values, in preparation for comparing simulation with hardware measurements in later experiments.

What is LTspice?

LTspice is a high-performance SPICE (Simulation Program with Integrated Circuit Emphasis) simulator, schematic capture tool, and waveform viewer, distributed free of charge by Analog Devices. It allows a circuit to be drawn on screen, assigned component values and a source, and then analysed numerically – computing node voltages and branch currents as functions of time (transient analysis), frequency (AC analysis), or DC bias (DC sweep/operating point analysis) – without building any physical hardware. Throughout this course, LTspice will be used *before* each hardware experiment to predict expected waveforms, and the simulated results will be compared with those measured on the DSO in the laboratory.

Basic Workflow

1. **Create a new schematic.** File → New Schematic, then save it with a suitable file-name before adding components (LTspice requires the file to be saved to run a simulation correctly).
2. **Place components.** Use the toolbar icons (or keyboard shortcuts) to place a resistor (R), capacitor (C), inductor (L), diode (D), ground (G), and a voltage source. Rotate/mirror using Ctrl+R/Ctrl+E.
3. **Wire the circuit.** Use F3 (or the wire tool) to connect component terminals. Every circuit *must* include a ground node (node 0), or the simulation will fail.
4. **Set component values.** Right-click a component and enter its value (e.g., 1k, 0.1u). For the source, right-click and choose the appropriate function (e.g., SINE for a sinusoidal source, specifying amplitude, frequency, and DC offset).
5. **Choose the analysis type.** Simulate → Edit Configure Analysis⁵:

⁵in Version 24.x

- `.tran` – transient (time-domain) analysis, for viewing waveforms vs. time.
 - `.ac` – AC sweep, for frequency response (Bode plots).
 - `.op` – DC operating point.
6. **Run the simulation.** Simulate $\rightarrow$ Run (or the RUN/PLAY icon). A waveform viewer window opens automatically.
 7. **Probe voltages and currents.** Click on a wire/node in the schematic to display its voltage, or click on a component to display the current through it, in the waveform viewer. To show the voltage between nodes, click and drag the probe between the nodes.
 8. **Add cursors and measure.** Use **Ctrl**+left-click on a trace label to attach a cursor; the cursor readout gives precise voltage, time, and (for two cursors) Δt and ΔV – directly analogous to DSO cursor measurements.

Pre-Lab Exercise

Using LTspice, draw the series RC network of Pre-Lab 1 (Figure 1) with the R and C values assigned to your batch:

- (a) Apply a **SINE** source with amplitude 1 V (peak) at $f = f_c$ (calculated in Pre-Lab 1) and run a `.tran` simulation for at least 5 cycles.
- (b) Using cursors, measure $V_{out,pp}$ and the time delay Δt between v_{in} and v_{out} ; hence compute ϕ using Equation (5), and compare with the theoretical value from Table 8.
- (c) Run a `.ac` simulation from $0.01f_c$ to $100f_c$ and plot the magnitude (in dB) and phase of v_{out}/v_{in} . Identify f_c on the plot as the -3 dB point.
- (d) Save screenshots of both simulations for submission along with your pre-lab record.

Pre-Lab Questions

- Q1. What is the difference between `.tran` and `.ac` analysis, and when would each be preferred?
- Q2. Why must every LTspice schematic contain a ground (node 0) reference?
- Q3. What is the purpose of the *maximum timestep* parameter in a `.tran` command, and what artefact can result if it is left too large?
- Q4. How would you model a real (non-ideal) function generator output impedance (typically $50\ \Omega$) in an LTspice source?

Resources

- LTspice download and official documentation (Analog Devices):
<https://www.analog.com/en/resources/design-tools-and-calculators/ltspice-simulator.html>
- LTspice Quick Reference / Getting Started guide: <https://www.analog.com/en/education/education-library/ltspice-getting-started-guides.html>
- Analog Devices LTspice video tutorial series (YouTube):
<https://www.youtube.com/@AnalogDevicesInc>
- LTwiki – community-maintained reference for LTspice syntax and models:
<https://ltwiki.org/>

Other free simulation tools

There are other SPICE-based free tools available which may also be used instead of LTspice. Some popular tools are listed here.

1. Qspice. Qspice is created by the creator of LTspice, Mike Engelhardt, as a modern, fast simulator: <https://p.qorvo.com/get-qspice>
2. Ngspice (requires a schematic capture tool): <https://ngspice.sourceforge.io>
3. eSim is an EDA package that goes beyond spice. Maintained by FOSSEE project under IITB. It uses ngspice as the simulation engine, and KiCad schematic for schematic entry. <https://esim.fossee.in/home/>

Chapter 1

Experiment 1: Clipping and Clamping Circuits using Diodes

1.1 Aim

- (i) To study the operation of series and shunt diode clipping circuits and to observe the effect of biasing on the clipping level.
- (ii) To study the operation of positive and negative diode clamping circuits.
- (iii) To design clipper and clamper circuits for specified output requirements.
- (iv) To verify, using a DSO, that the observed waveforms agree with the theoretically predicted transfer characteristics.

1.2 Pre-Lab Exercise

Before performing the experiment, simulate the circuits and verify the designs using LTspice or any other circuit simulation software.

1.3 Equipment/Components Required

S.No.	Equipment/Component	Specification
1	Digital Storage Oscilloscope	2-channel, 20–50 MHz bandwidth
2	Function Generator ¹	Sine wave, 1 Hz – 1 MHz
3	Regulated DC Power Supply	0–30 V
4	Silicon diode	1N4007 (or equivalent), 2 nos.
5	Zener diodes	$V_z = 5.6$ V, 2 nos.
6	Resistors	1 k Ω – 10 k Ω
7	Capacitors	1 μ F – 10 μ F
8	Breadboard and connecting wires	–

Table 1.1: Equipment and component list.

1.4 Theory

Diode clipping and clamping circuits exploit the nonlinear, near-ideal-switch behaviour of a p – n junction diode to reshape a waveform. A *clipper* removes (clips) a portion of the input waveform

above and/or below a reference level, while a *clammer* shifts the entire waveform up or down without altering its shape, thereby changing its DC reference level.

Throughout this document, the diode is modelled as an ideal switch in series with a cut-in voltage $V_F \approx 0.7\text{ V}$ (silicon) and a forward resistance $R_f \approx 0$; the reverse resistance is assumed infinite.

1.4.1 Clipping Circuits

(a) Series (Half-Wave) Clipper. Figure 1.1 shows a series diode clipper. The diode conducts only when v_{in} forward-biases it, so the circuit passes only one half-cycle of the input to the output. The output voltage across the resistance R_L will be $v_{in}(t) - V_F$.

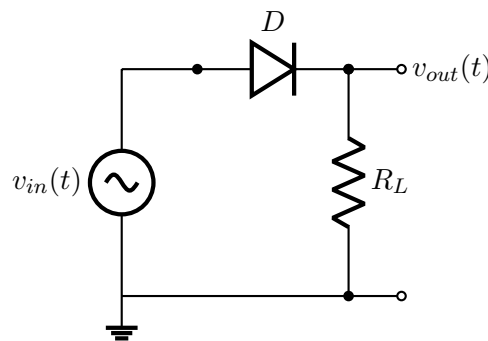


Figure 1.1: Series (half-wave) diode clipper. The diode conducts on the positive half-cycle and passes it to R_L ; the negative half-cycle is clipped.

(b) Shunt (Parallel) Clipper – Positive Clipper. Figure 1.2a shows a shunt clipper with the diode across the output. Whenever $v_{in} > V_F$, the diode conducts and clamps the output near V_F ; for $v_{in} < V_F$, the diode is OFF and $v_{out} = v_{in} \frac{R_L}{R_L + R}$. If $R_L \gg R$, $v_{out} \approx v_{in}$. The waveforms expected are as given in Fig. 1.2b.

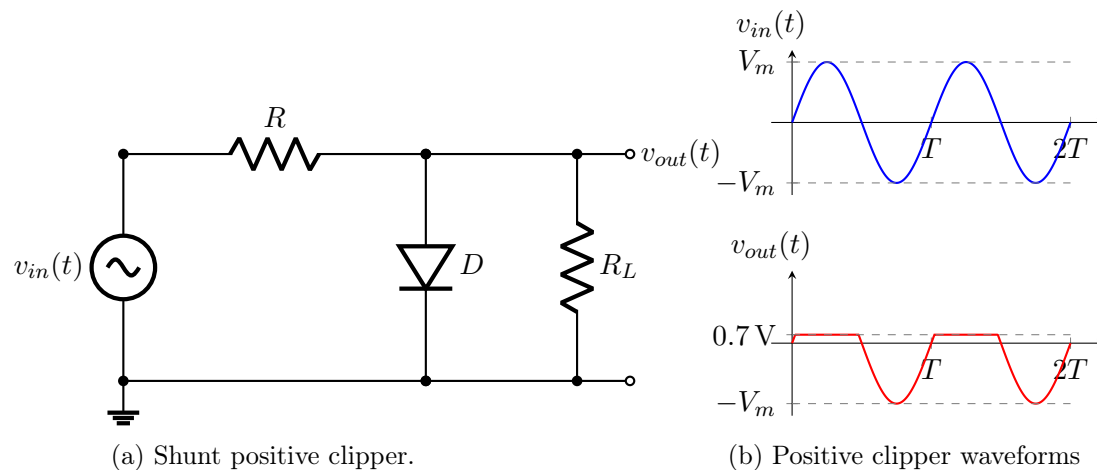


Figure 1.2: Diode clipper circuit with input and output waveforms

The positive portion of the input waveform is clipped off, hence the name positive clipper.

(c) Shunt (Parallel) Clipper – Negative Clipper. Figure 1.3a shows a shunt negative clipper with the diode across the output, placed in the opposite polarity compared to the positive clipper. Whenever $v_{in} < -V_F$, the diode conducts and clamps the output near $-V_F$; for $v_{in} >$

$-V_F$, the diode is OFF and $v_{out} = v_{in} \frac{R_L}{R_L + R}$. If $R_L \gg R$, $v_{out} \approx v_{in}$. The waveforms expected are as given in Fig. 1.3b.

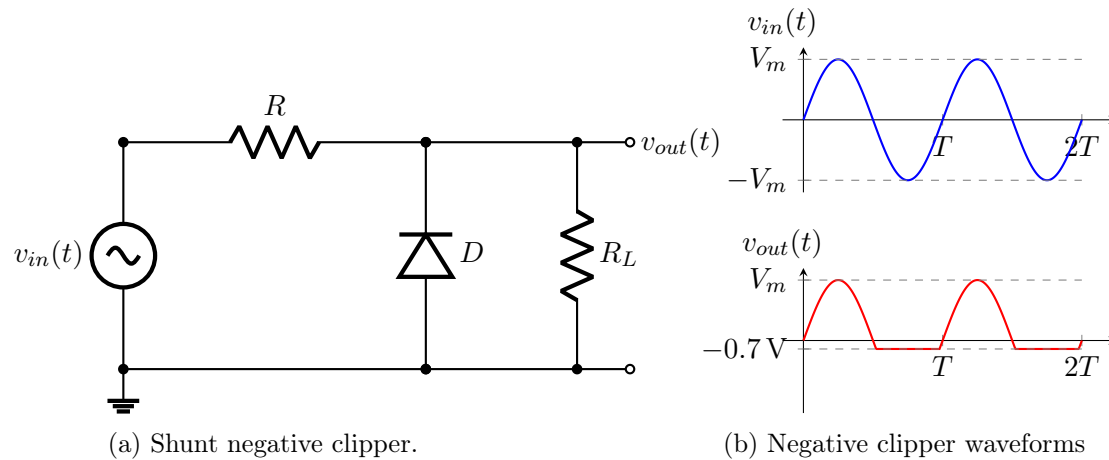


Figure 1.3: Diode clipper circuit with input and output waveforms

The negative portion of the input waveform is clipped off, hence the name negative clipper.

(d) Biased Clippers. A DC bias source V_R in series with the diode (Figure 1.4) shifts the clipping level from V_F to $(V_R + V_F)$ or $(V_R - V_F)$, depending on polarity. This allows the clipping level to be set anywhere on the waveform. There are two ways to achieve the biasing. One is to provide a direct dc source. The other is to use a Zener diode or series combination of diodes to achieve the bias level.

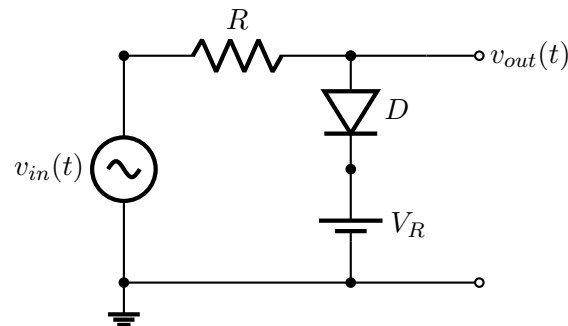


Figure 1.4: Biased shunt clipper. Output is clipped at approximately $V_R + V_F$.

Fig. 1.5a shows a biased positive clipper with a dc bias. Fig. 1.5b shows the implementation with a Zener diode. The clipping levels are $V_R + V_F$ and $V_Z + V_F$ respectively. Fig. 1.6 shows the sample waveforms.

In a similar way, a **Biased Negative Clipper** can be implemented by reversing the diode polarity. The figures 1.7a and 1.7b show the circuit implementations.

The sample waveforms for the biased negative clipper are as shown in Fig. 1.8.

(Double (Two-Level) Clipper or Slicer. Two diodes, oppositely poled and biased at V_{R1} and V_{R2} , in parallel (Figure 1.9a), clip both the positive peak above $(V_{R1} + V_F)$ and the negative peak below $-(V_{R2} + V_F)$, producing a trapezoidal output for a sinusoidal input of sufficient amplitude. This circuit is also known as a ‘slicer’.

Instead of DC biasing voltages, zener diodes can be used.

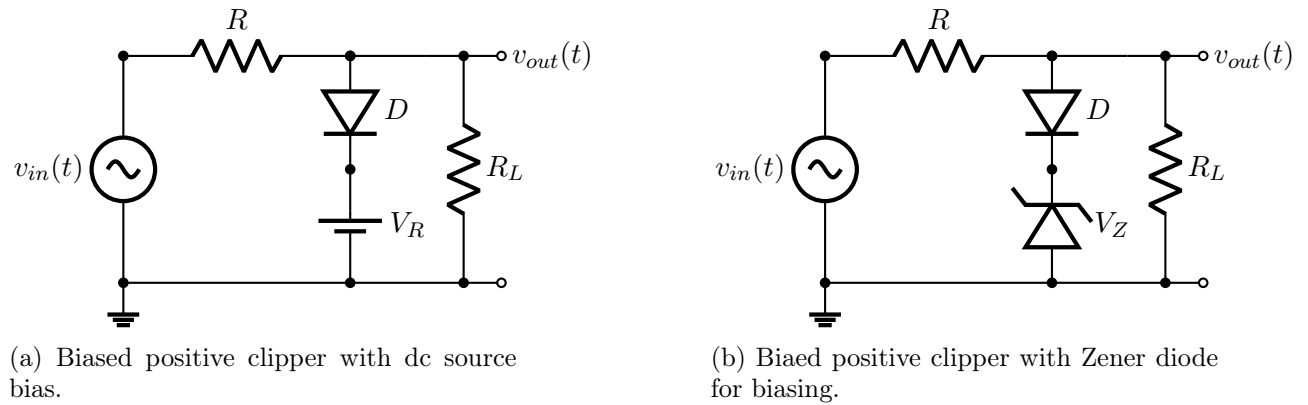
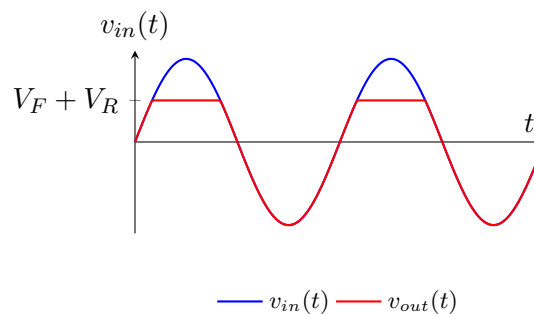


Figure 1.5: Biased Positive Clipper Configurations


 Figure 1.6: Expected input and output waveforms for the biased positive clipper of Figure 1.5a: the output is flattened above V_F .

1.4.2 Clamping Circuits

A clamper consists of a diode, a capacitor, and a resistor, arranged so that the capacitor charges (through the low forward resistance of the diode) to the peak of the input during one half-cycle and holds this charge (discharging slowly through R_L , with $R_L C \gg T$) during the rest of the cycle, thereby shifting the DC level of the waveform. It essentially ‘clamps’ the input signal up or down by the peak voltage. It is also called a DC restorer.

Mathematically, for an input signal given by $v_{in}(t) = V_m \sin(\omega t)$, the clamper output is $v_o(t) = \pm V_m + V_m \sin(\omega t)$.

(a) Negative Clamper. Figure 1.10a shows a negative clamper, which shifts the waveform downward so that its positive peak is clamped at (approximately) 0 V (or at V_F , accounting for diode drop).

(b) Positive Clamper. Reversing the diode polarity (Figure 1.11a) clamps the negative peak of the waveform at approximately 0 V, shifting the entire waveform upward.

For an input sine wave of amplitude V_m , the steady-state clamped output ideally swings between:

$$\text{Negative clamper:} \quad v_{out} \in [-2V_m + V_F, V_F], \quad (1.1)$$

$$\text{Positive clamper:} \quad v_{out} \in [-V_F, 2V_m - V_F]. \quad (1.2)$$

The time constant $R_L C$ must satisfy $R_L C \gg T$ (the input period) so that the capacitor voltage does not decay appreciably between successive charging peaks (poor choice of $R_L C$ leads to *waveform tilt/sag*).

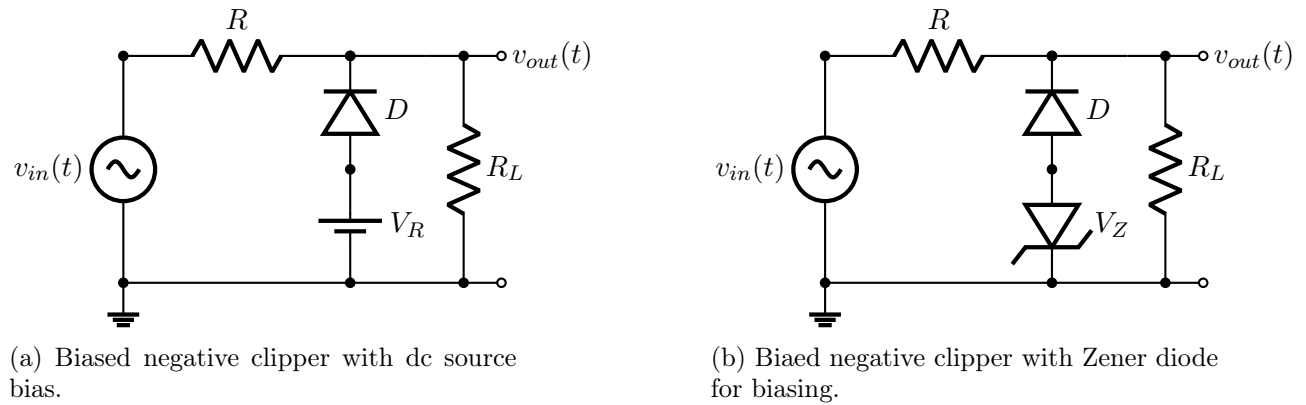


Figure 1.7: Biased Negative Clipper Implementations

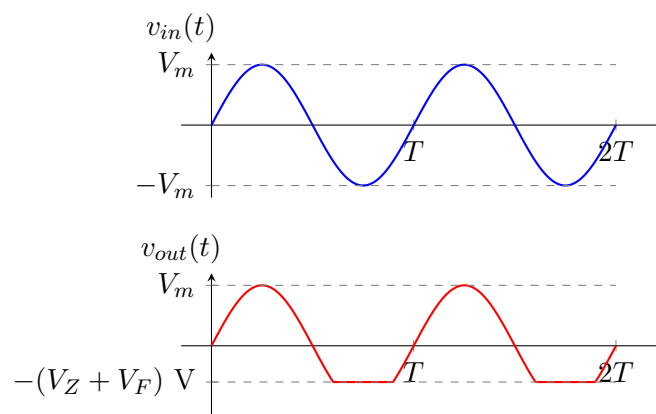


Figure 1.8: Biased Negative Clipper Waveforms (For Zener based biasing)

1.5 Circuit Diagrams

The circuits are as shown in the previous sections.

Note

While preparing the lab-report, circuit diagrams shall contain proper labels for the components, including the component part numbers (such as 1N4001), and values/parameters (example: $5 V_{pp}$, Sine, 50 Hz, for the source; R_1 , 50 k Ω , etc.).

1.6 Design Procedure

- Select the operating signal's frequency and amplitude, V_m and f .
- For a biased clipper required to clip at a level V_{clip} , select $V_R = V_{clip} - V_F$ using a potential divider from the DC supply, or a battery/regulated source directly. A Zener diode may also be chosen, as described earlier.
- For a clamper, choose C such that $R_L C \geq 10T$, where $T = 1/f$ is the period of the lowest input frequency to be used, so that the sag in output during discharge is limited to within an acceptable fraction (typically $< 5\%$) of V_m .
- Choose R_L large enough to limit diode conduction current to a safe value, but small enough to keep $R_L C \gg T$ practical with standard capacitor values.

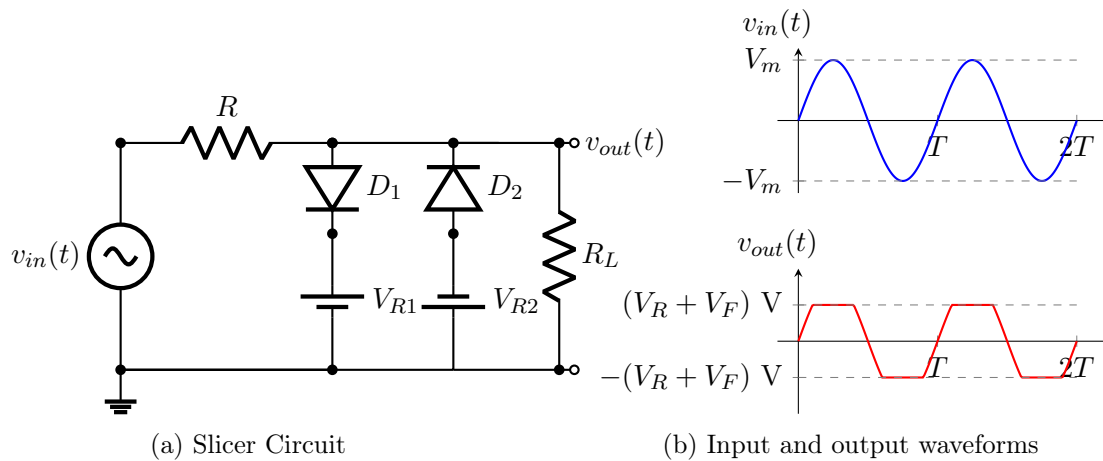


Figure 1.9: Slicer circuit and waveforms

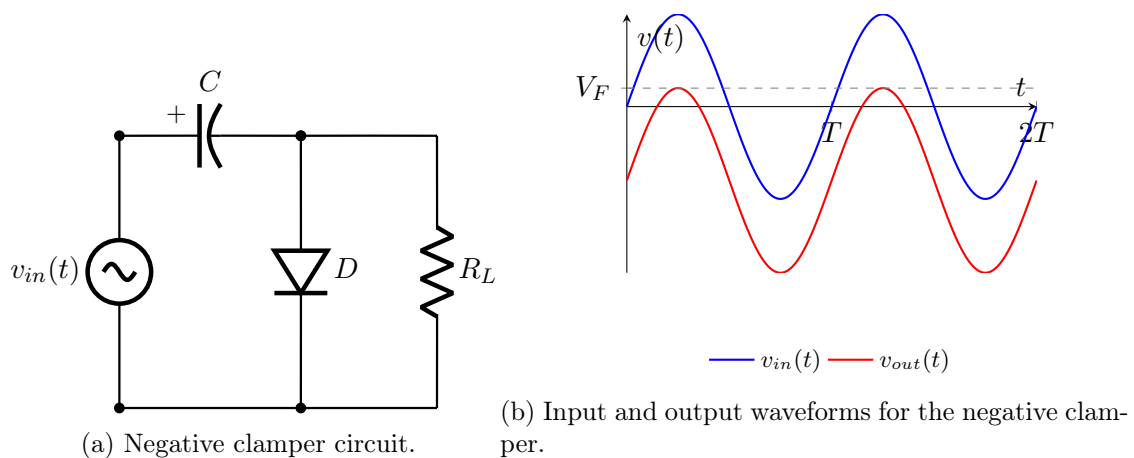
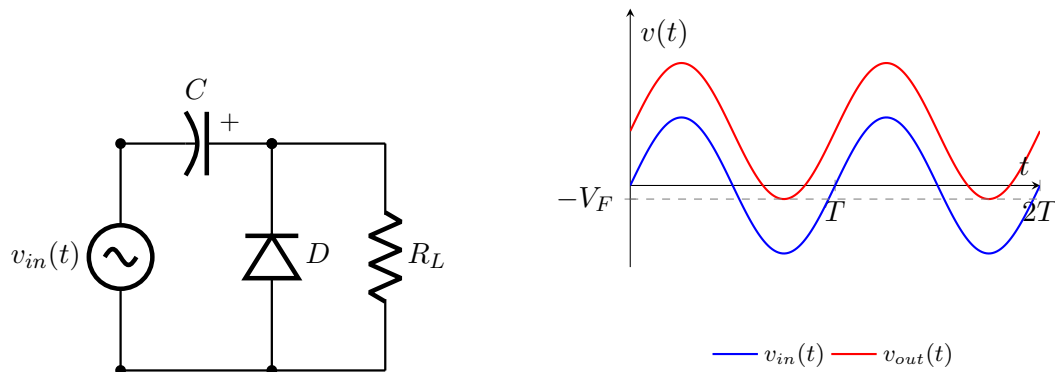


Figure 1.10: Negative Clamper: Circuit and waveforms

- (e) **Selection of diodes:** For input frequencies less than 1 kHz, line frequency diodes such as 1N4001 may be used. For higher frequencies, these slow diodes are to be replaced with faster signal diodes such as 1N4148.

1.7 Experiment Procedure

1. Assemble the shunt positive clipper of Figure 1.2a on the breadboard. Set the function generator to a sine wave of frequency 1 kHz and amplitude $5 V_{pp}$.
2. Connect DSO CH1 to v_{in} and CH2 to v_{out} , with a common ground. Display both waveforms simultaneously and sketch/record them.
3. Replace the shunt positive clipper with the shunt negative clipper. (Figure 1.3a). Record v_{in} and v_{out} , and measure the clipping level from the DSO screen using the horizontal cursors.
4. Repeat for the biased clipper (Figure 1.4) for at least two values of V_R , and the double clipper (Figure 1.9a).
5. For each clipper configuration, tabulate the theoretically expected and experimentally measured clipping levels.



(a) Positive clamper circuit (diode polarity reversed with respect to Figure 1.10a). (b) Input and output waveforms for the negative clamper.

Figure 1.11: Positive clamper: Circuit and waveforms

6. Assemble the negative clamper (Figure 1.10a) using the design values of R_L and C computed in the Design Procedure. Apply the same sine wave and record v_{in} and v_{out} on the DSO, noting the DC shift and any tilt/sag in the output.
7. Repeat for the positive clamper (Figure 1.11a).
8. For each clamper, measure $V_{out,max}$ and $V_{out,min}$ using DSO cursors and compare with Equations (1.1)–(1.2).
9. Vary the input frequency (e.g., to 100 Hz) while keeping $R_L C$ fixed, and observe/record the effect on clamper output tilt. Explain the observation.

1.8 Precautions

- Observe correct diode polarity while assembling each circuit; an inverted diode produces the mirror-image (and often misleading) result.
- Do not exceed the diode's maximum forward current or reverse voltage ratings.
- Ensure a common ground between the function generator and both DSO channels.
- For clamping circuits, allow a few cycles for the capacitor to reach steady-state charge before recording the output.
- Use electrolytic capacitors, if used, with correct polarity as per the expected DC level at that node.

1.9 Expected Waveforms

The sample waveforms for the clipping circuits are as shown in Figs. 1.2b, 1.3b, 1.6, and 1.9b. The waveforms for the clamping circuits are as shown in Figs. 1.10b, 1.11b. When recording waveforms, note down all the salient points/features of the waveforms, such as voltage/time details.

1.10 Observation Tables

The observation table for the clipping circuits is given in Table 1.2. Use Table 1.3 to tabulate the Clamper circuit observations.

Circuit	V_{clip} (theory)	V_{clip} (measured)	% Error
Shunt positive clipper			
Shunt negative clipper			
Biased positive clipper (V_{R1})			
Biased negative clipper (V_Z)			
Double clipper (upper level)			
Double clipper (lower level)			

Table 1.2: Clipping circuits: theoretical vs. measured clipping levels.

Circuit	$V_{out,max}$ (theory)	$V_{out,max}$ (meas.)	$V_{out,min}$ (theory)	$V_{out,min}$ (meas.)
Negative clamper				
Positive clamper				

Table 1.3: Clamping circuits: theoretical vs. measured output extremes.

1.11 Results

1. The diode-based clipping circuits with and without external bias are set-up and the output waveforms were studied.
2. The diode-based clamping circuits were set-up and the waveforms were studied.

Note

Experiment with higher frequency signals. What do you observe? Can you explain why circuit performance is not as expected at higher frequencies?

1.11.1 Questions

- Q1. Distinguish between series and shunt clipper configurations in terms of output impedance and loading effects.
- Q2. Derive the expression for the clipping level of the biased clipper of Figure 1.4 in terms of V_R and V_F .
- Q3. Why must the time constant $R_L C$ of a clamper be much greater than the period of the input waveform? What happens if $R_L C$ is comparable to T ?
- Q4. Sketch the output waveform of the double clipper (Figure 1.9a) for an input triangular wave of amplitude greater than $(V_{R1} + V_{R2} + 2V_F)$.
- Q5. A clamper is required to shift a $10 V_{pp}$, 1 kHz sine wave so that its output never goes negative. Design suitable values of R_L and C , assuming a maximum permissible sag of 2%.
- Q6. Explain why a clipper is sometimes called a *limiter* and give one practical application each of a clipper and a clamper in electronic systems.
- Q7. Why do clipper circuits do not perform as intended when diodes like 1N4001 are used, at higher frequencies, such as 50 kHz?

Chapter 2

Experiment 2: Zener Diode Voltage Regulator

2.1 Aim

To design, and construct a Zener diode based DC voltage regulator for a given load current and input voltage range, and to study its line regulation and load regulation characteristics.

2.2 Pre-Lab Exercise

Before performing the experiment, simulate the circuits and verify the designs using LTspice or any other circuit simulation software.

2.3 Apparatus / Components Required

S.No.	Component	Specification
1	Zener diode	$V_Z = 5.6 \text{ V}$, $P_{Z(\max)} = 400 \text{ mW}$
2	Resistor R_S	As designed (series/current-limiting resistor)
3	Resistor (as R_L)	1–10 k Ω
4	Regulated DC Power Supply	0–15 V, variable
5	Digital Multimeter	–
6	Breadboard and connecting wires	–

2.4 Theory

A Zener diode, when reverse biased beyond its breakdown voltage V_Z , maintains a nearly constant voltage across its terminals over a wide range of reverse current. This property is exploited to build a simple shunt voltage regulator: a series resistor R_S limits the current drawn from the unregulated input V_{in} , while the Zener diode, connected in parallel with the load R_L , clamps the output voltage to V_Z as long as it is operated between its minimum breakdown-sustaining current $I_{Z(\min)}$ and its maximum rated current $I_{Z(\max)}$ (set by its power rating).

The regulator must continue to hold $V_{out} \approx V_Z$ under two independent disturbances:

- **Line regulation:** V_{in} varies over its specified range while I_L is held constant.
- **Load regulation:** I_L varies (via R_L) while V_{in} is held constant.

2.5 Design

Given specifications.

$$\begin{aligned}V_Z &= 5.6 \text{ V} \\I_{L(\max)} &= 5 \text{ mA} \\V_{in} &: 8 \text{ V to } 15 \text{ V} \quad (V_{in(\min)} = 8 \text{ V}, V_{in(\max)} = 15 \text{ V}) \\P_{Z(\max)} &= 400 \text{ mW}\end{aligned}$$

Step 1: Maximum Zener current from the power rating.

$$I_{Z(\max)} = \frac{P_{Z(\max)}}{V_Z} = \frac{400 \text{ mW}}{5.6 \text{ V}} \approx 71.4 \text{ mA}$$

Although this is the maximum Zener current, to operate without damage, a much lower current is assumed. Hence, an $I_{Z(max)}$ is assumed to be 50 mA.

Step 2: Choose minimum Zener current. To ensure the diode stays in the breakdown region even at worst case (minimum V_{in} , maximum I_L), a minimum holding current is chosen as

$$I_{Z(\min)} = 5 \text{ mA} \quad (\text{a conservative, commonly used design margin})$$

Step 3: Design R_S for the worst case (minimum V_{in} , maximum I_L). At $V_{in(\min)}$, the source must supply both $I_{L(\max)}$ and $I_{Z(\min)}$:

$$R_S = \frac{V_{in(\min)} - V_Z}{I_{L(\max)} + I_{Z(\min)}} = \frac{8 - 5.6}{(5 + 5) \times 10^{-3}} = \frac{2.4}{0.010} = 240 \, \Omega$$

Choosing the nearest standard (E12) value: $R_S = 220 \, \Omega$.

Step 4: Verify at minimum V_{in} , full load (checks regulation is maintained).

$$I_T = \frac{V_{in(\min)} - V_Z}{R_S} = \frac{2.4}{220} \approx 10.9 \text{ mA}$$

$$I_Z = I_T - I_{L(\max)} = 10.9 - 5 = 5.9 \text{ mA} \quad (> I_{Z(\min)} = 5 \text{ mA}) \quad \checkmark$$

Step 5: Verify at maximum V_{in} , no load (checks Zener power rating is not exceeded).

$$I_Z = \frac{V_{in(\max)} - V_Z}{R_S} = \frac{15 - 5.6}{220} \approx 42.7 \text{ mA}$$

$$P_Z = I_Z \times V_Z = 42.7 \text{ mA} \times 5.6 \text{ V} \approx 239 \text{ mW} \quad (< 400 \text{ mW}) \quad \checkmark$$

Step 6: Power rating of R_S .

$$P_{R_S} = \frac{(V_{in(\max)} - V_Z)^2}{R_S} = \frac{(9.4)^2}{220} \approx 0.40 \text{ W}$$

Use a $220 \, \Omega$, 1 W resistor (adequate safety margin over the calculated 0.40 W).

Final design summary.

Parameter	Value
Zener diode	5.6 V, 400 mW
Series resistor R_S	220 Ω , 1 W
Load current range	0 to 5 mA
Input voltage range	8 V to 15 V
Regulated output	≈ 5.6 V

2.6 Circuit Diagram

The circuit diagram to be set up is given in Fig. 2.1.

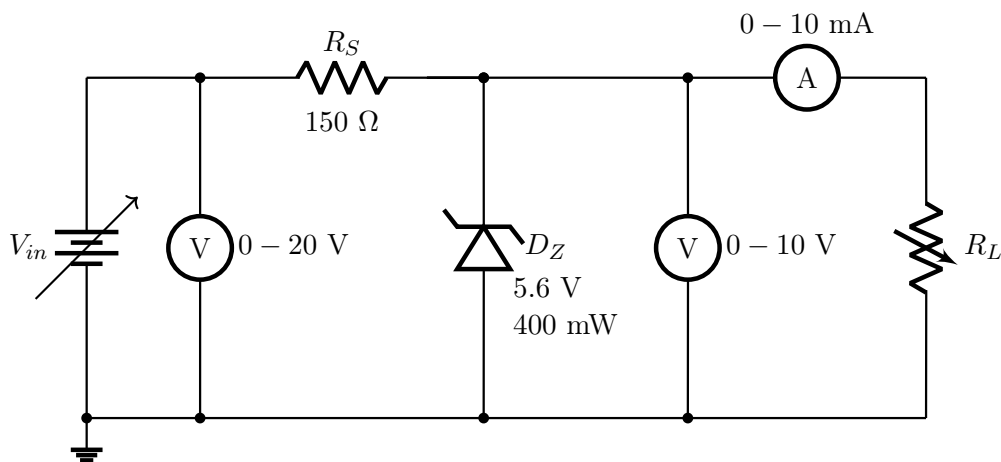


Figure 2.1: Zener diode voltage regulator circuit

2.7 Procedure**(a) Line regulation.**

1. Connect the circuit as shown in Figure 2.1 with R_L set to give $I_L = I_{L(\max)} = 5$ mA (i.e., $R_L = V_Z/I_{L(\max)} \approx 1.12$ k Ω). A rheostat of 2.2k may be used as the load resistance R_L .
2. Vary V_{in} in steps from 8 V to 15 V, keeping R_L fixed.
3. Note V_{out} across the Zener/load for each value of V_{in} .
4. Plot V_{out} versus V_{in} .

(b) Load regulation.

1. Fix V_{in} at a convenient value within the design range (e.g., $V_{in} = 12$ V).
2. Vary R_L (using the decade resistance box) so that I_L varies from 0 (open circuit) to $I_{L(\max)} = 5$ mA.
3. Note V_{out} for each value of I_L .
4. Plot V_{out} versus I_L .

2.8 Tabulation

Line regulation (I_L held at 5 mA).

V_{in} (V)	V_{out} (V)
8	
9	
10	
11	
12	
13	
14	
15	

Load regulation (V_{in} held constant).

I_L (mA)	V_{out} (V)
0	
1	
2	
3	
4	
5	

2.9 Model Graph

Plot two graphs from the observations:

1. V_{out} (Y-axis) vs. V_{in} (X-axis) — should be nearly flat once V_{in} exceeds the value needed to bring the Zener into breakdown, confirming line regulation.
2. V_{out} (Y-axis) vs. I_L (X-axis) — should remain close to V_Z up to $I_{L(\max)}$, with a slight droop as I_L approaches the design limit, confirming load regulation.

2.10 Calculations from Observations

$$\% \text{ Line regulation} = \frac{\Delta V_{out}}{\Delta V_{in}} \times 100\%$$

$$\% \text{ Load regulation} = \frac{V_{out(\text{no load})} - V_{out(\text{full load})}}{V_{out(\text{full load})}} \times 100\%$$

2.11 Result

A Zener diode voltage regulator was designed for $V_Z = 5.6$ V, $I_{L(\max)} = 5$ mA, over an input range of 8–15 V, using a 400 mW Zener diode and a series resistor $R_S = 220 \Omega$. The circuit was constructed and its line regulation and load regulation characteristics were verified experimentally.

The maximum load regulation observed = _____, at $I_L =$ _____

The maximum line-regulation observed = _____, at $V_{in} =$ _____

2.12 Questions

1. Why is the Zener diode operated in reverse breakdown for voltage regulation?
2. What determines the maximum and minimum permissible Zener current?
3. Why must R_S be designed using the minimum input voltage and maximum load current condition?
4. What happens to the regulator if R_L is accidentally open-circuited?
5. Define line regulation and load regulation. What is an ideal value for each?
6. How does the power rating of the Zener diode limit the input voltage range for a fixed R_S ?

Chapter 3

Experiment 3: Basic RC Circuits - High-pass and Low-pass Filters

3.1 Aim

To design, construct and study the frequency response (Bode plot) and time-domain behaviour of a first-order RC low-pass filter and a first-order RC high-pass filter for a specified cutoff frequency.

3.2 Pre-Lab Exercise

Before performing the experiment, simulate the circuits and verify the designs using LTspice or any other circuit simulation software.

3.3 Apparatus / Components Required

S.No.	Component	Specification
1	Resistor R	As designed
2	Capacitor C	As designed
3	Function Generator	Sine and square wave, up to 100 kHz
4	Cathode Ray Oscilloscope (CRO)	Dual channel
5	Breadboard and connecting wires	–

3.4 Theory

We have seen the RC circuit in the pre-lab activity, where we used this circuit to familiarise the DSO and the function generator and the measurement techniques.

A first-order RC low-pass filter (LPF) passes low frequencies and attenuates (reduces the amplitude of) high frequencies, while a first-order RC high-pass filter (HPF) does the opposite. Both are built from the same two components, only the position of R and C is interchanged, and both share the same cutoff (or corner) frequency

$$f_c = \frac{1}{2\pi RC}.$$

The attenuation/gain is measured in terms of the unit decibel (dB), where a dB is defined as follows:

$$dB = 20 \log_{10} \frac{V_{out}}{V_{in}}$$

Low-pass filter. With R in series and C shunting the output to ground, the transfer function is

$$H_{LP}(f) = \frac{V_{out}}{V_{in}} = \frac{1}{1 + j(f/f_c)}, \quad |H_{LP}(f)| = \frac{1}{\sqrt{1 + (f/f_c)^2}}, \quad \angle H_{LP}(f) = -\tan^{-1}\left(\frac{f}{f_c}\right).$$

High-pass filter. With C in series and R shunting the output to ground, the transfer function is

$$H_{HP}(f) = \frac{V_{out}}{V_{in}} = \frac{j(f/f_c)}{1 + j(f/f_c)}, \quad |H_{HP}(f)| = \frac{f/f_c}{\sqrt{1 + (f/f_c)^2}}, \quad \angle H_{HP}(f) = 90^\circ - \tan^{-1}\left(\frac{f}{f_c}\right).$$

At $f = f_c$, both filters attenuate the signal to $1/\sqrt{2}$ (−3 dB) of the input, with a phase shift of -45° (LPF) or $+45^\circ$ (HPF).

Time-domain view. When the input is a square wave:

- If $RC \gg T$ (period), the LPF cannot follow the fast transitions and behaves like an *integrator*, producing a triangular/rounded output.
- If $RC \ll T$, the HPF passes only the transitions and behaves like a *differentiator*, producing narrow decaying spikes at each edge.

3.5 Design

Specifications.

$$f_c = 1 \text{ kHz}$$

Step 1: Choose a standard capacitor value.

$$C = 0.1 \text{ } \mu\text{F} \quad (\text{standard value})$$

Step 2: Compute R for the required cutoff frequency.

$$R = \frac{1}{2\pi f_c C} = \frac{1}{2\pi \times 1000 \times 0.1 \times 10^{-6}} \approx 1591.5 \text{ } \Omega$$

Nearest standard value: $R = 1.5 \text{ k}\Omega$.

Step 3: Verify actual cutoff frequency.

$$f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi \times 1500 \times 0.1 \times 10^{-6}} \approx 1.061 \times 10^3 \text{ Hz} \approx 1 \text{ kHz} \quad (\text{OK})$$

Final design summary (common to both filters).

Parameter	Value
Resistor R	1.5 k Ω
Capacitor C	0.1 μF
Cutoff frequency f_c	$\approx 1 \text{ kHz}$
Attenuation at f_c	−3 dB ($0.707 \times V_{in}$)
Phase shift at f_c	−45° (LPF), +45° (HPF)

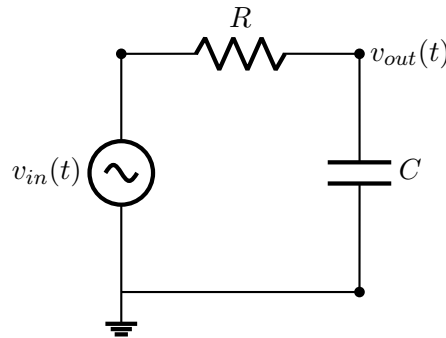


Figure 3.1: First-order RC low-pass filter

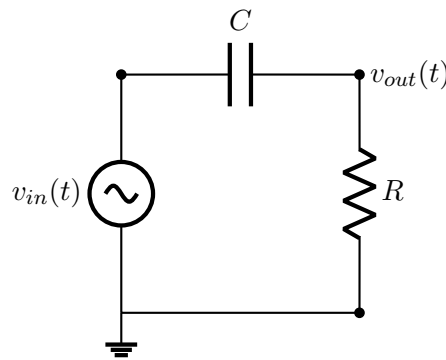


Figure 3.2: First-order RC high-pass filter

3.6 Circuit Diagrams

The circuits of the RC low-pass and high-pass filters are shown in Fig. 3.1 and Fig. 3.2 respectively.

3.7 Procedure

(a) Frequency response (Bode plot).

1. Connect the circuit as shown in Figure 3.1 (repeat later for Figure 3.2).
2. Apply a sine wave of constant amplitude V_{in} from the function generator.
3. Vary the frequency from about 10 Hz to 100 kHz in suitable steps (more closely spaced near f_c).
4. At each frequency, measure V_{out} and the phase difference between $v_{in}(t)$ and $v_{out}(t)$ using the CRO (dual trace / Lissajous method).
5. Compute gain in dB as $20 \log_{10}(V_{out}/V_{in})$. Tabulate the observations/calculations as in Table ??.
6. Plot gain (dB) and phase (degrees) versus frequency (log scale) — the Bode magnitude and phase plots. Use a semi-log graph-sheet for plotting the frequency response.
7. Gain (dB) versus frequency for the LPF and HPF, each expected to approach the shapes shown in Figures 3.3 and 3.4, with a -3 dB point at f_c and asymptotic slopes of ± 20 dB/decade away from f_c .
8. Phase (degrees) versus frequency for the LPF and HPF, passing through -45° and $+45^\circ$ respectively at f_c .

(b) Time-domain response.

1. Apply a square wave at a frequency well below f_c (for the LPF, to observe near-integrator behaviour) and well above f_c (for the HPF, to observe near-differentiator behaviour).
2. Observe and sketch $v_{in}(t)$ and $v_{out}(t)$ simultaneously on the CRO.
3. Compare the observed waveforms with Figures 3.5 and 3.6.

3.8 Tabulation**Low-pass filter.**

Frequency (Hz)	V_{out} (V)	Gain = $20 \log_{10}(V_{out}/V_{in})$ (dB)	Phase (deg)
10			
100			
500			
1000			
2000			
5000			
10000			
50000			

High-pass filter.

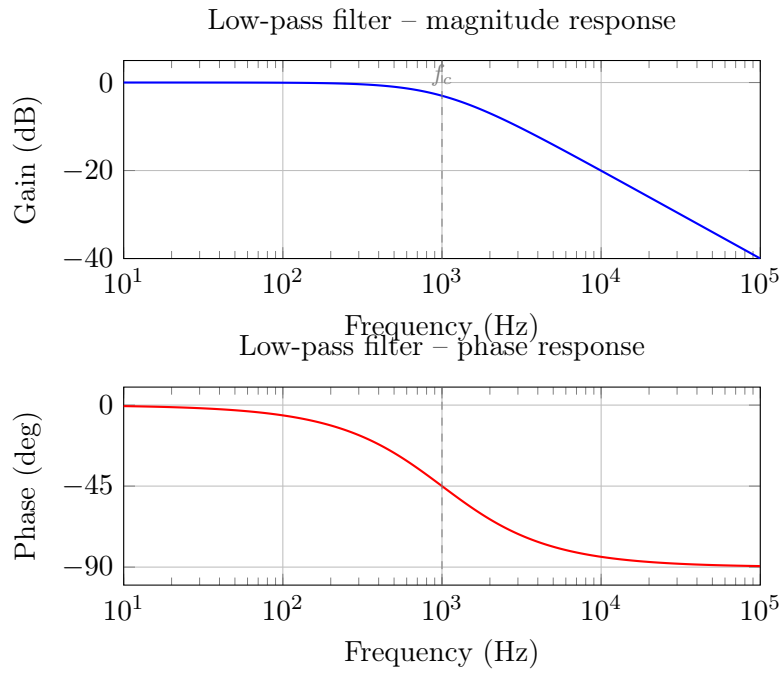
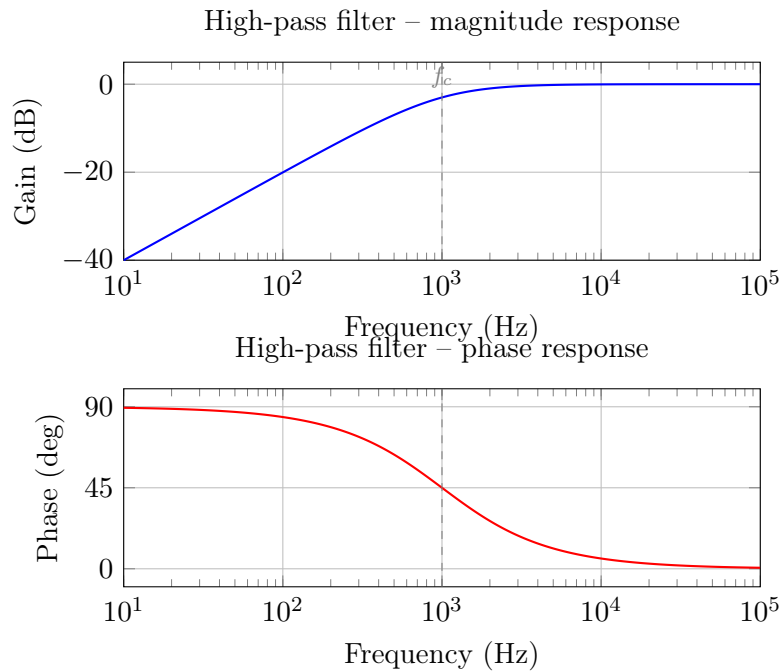
Frequency (Hz)	V_{out} (V)	Gain = $20 \log_{10}(V_{out}/V_{in})$ (dB)	Phase (deg)
10			
100			
500			
1000			
2000			
5000			
10000			
50000			

Bode Diagrams

The frequency-response curves (gain and phase Vs frequency) are known as Bode diagrams (after the inventor Hendrik Wade Bode). The y-axis of the gain plot uses the dB unit, and the phase plot uses the phase lag/lead in degrees. The x-axis is in log-scale, and frequencies in decades are marked. The sample Bode diagrams are as shown in Figs. 3.3 and 3.4.

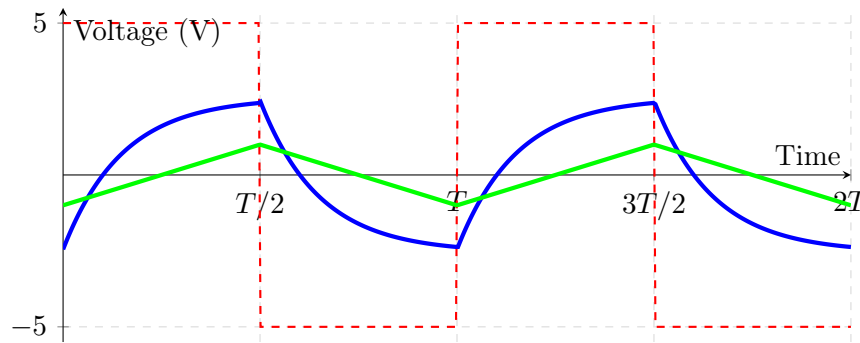
3.9 Sample Waveforms (Square Wave Input)

The time-domain behaviour of the RC circuit can be understood by studying the response to a square-wave of variable frequency, the waveforms of which are shown in Figs. 3.5 and 3.6. It can be seen that for $RC \gg T/2$, the RC low-pass filter circuit acts as an integrator, whereas for the RC high-pass circuit (with C and R interchanged), behaves as a differentiator when $RC \ll T/2$. Further, note that the waveform given in Fig. 3.6 are obtained under the constraint that $R \gg R_{out}$ of the source (function generator).

Figure 3.3: Bode plot of the RC low-pass filter ($f_c \approx 1$ kHz)Figure 3.4: Bode plot of the RC high-pass filter ($f_c \approx 1$ kHz)

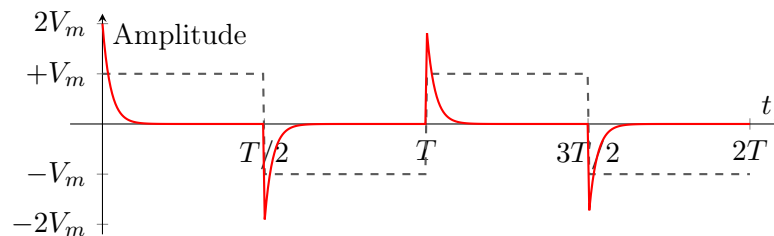
3.10 Result

First-order RC low-pass and high-pass filters were designed for a cutoff frequency of 1 kHz using $R = 1.5$ k Ω and $C = 0.1$ μ F. Their frequency response (Bode plots) and time-domain behaviour with a square-wave input were studied and verified experimentally against the theoretical response.



--- Input V_{in} — Output V_{out} for $RC < T$ — Output V_{out} for $RC \gg T$

Figure 3.5: Low-pass filter response to a square wave (rounded/ramped output)



--- Input square wave — HPF output ($RC \ll T$: differentiator)

Figure 3.6: High-pass filter response to a square wave (decaying spikes at transitions)

Explore further

The output impedance of the source (function generator) has an impact on practical measurements. Hence, while checking the output waveforms of the high-pass (or differentiator) circuit, the resistance value should be much higher than that of the output impedance of the function generator (typically, $50\ \Omega$). For satisfying the condition $RC \ll T/2$, we may choose a higher R and a lower C , or vice-versa. Try with both designs and see how the waveforms differ in each case practically, and explain why. What is the effect in the case of the integrator? When simulating, you may experiment with/without a source resistance to see the effects.

3.11 Questions

1. Derive the transfer function of a first-order RC low-pass filter.
2. Why do the LPF and HPF built from the same R and C have the same cutoff frequency?
3. What is the significance of the -3 dB point?
4. What is the asymptotic roll-off rate of a first-order filter, in dB/decade?
5. Under what condition does an RC low-pass filter behave as an integrator, and a high-pass filter as a differentiator?
6. Why is the RC circuit (both LPF and HPF) in this experiment called “first-order” filters?

7. How would you extend this circuit to a second-order filter, and what advantage would that give?
8. What are the effects of function generator output impedance on the design/expected waveforms of the RC circuit?
9. How do the integrator and differentiator circuits behave when the input signal has a DC added to it ($\pm V_{dc}$)?